

Email spam detection -2

```
from google.colab import drive  
drive.mount("/content/drive")
```

```
import os  
os.listdir("/content/drive/MyDrive")
```

```
!pip install kagglehub  
import kagglehub
```

```
path = kagglehub.dataset_download("uciml/sms-spam-collection-dataset")  
print("Path to dataset files:", path)
```

```
import os

import pandas as pd

# Check which files are inside the dataset folder

print(os.listdir("/root/.cache/kagglehub/datasets/uciml/sms-spam-collection-
dataset/versions/1"))

# Load the dataset

df = pd.read_csv("/root/.cache/kagglehub/datasets/uciml/sms-spam-collection-
dataset/versions/1/spam.csv", encoding='latin-1')

# Show first few rows

df.head()
```

```
import pandas as pd

# Keep only the required columns and rename them
df = df.rename(columns={'v1': 'label', 'v2': 'message'})
df = df[['label', 'message']]
```

```
# Display cleaned dataset
df.head()
```

```
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.model_selection import train_test_split
```

```
# Convert text to numeric vectors
vectorizer = TfidfVectorizer(stop_words='english')
X = vectorizer.fit_transform(df['message'])
y = df['label'].map({'ham': 0, 'spam': 1}) # 0=ham, 1=spam
```

```
# Split dataset
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)
```

```
from sklearn.svm import SVC

from sklearn.metrics import accuracy_score, confusion_matrix


# Train model

model = SVC(kernel='linear')

model.fit(X_train, y_train)

y_pred = model.predict(X_test)


# Show accuracy and confusion matrix

print("Accuracy:", accuracy_score(y_test, y_pred))

print("Confusion Matrix:\n", confusion_matrix(y_test, y_pred))
```

```
sample = [

    "You have won $1000! Claim now!",

    "Hey, are we meeting tomorrow?"

]

sample_features = vectorizer.transform(sample)

pred = model.predict(sample_features)


for msg, p in zip(sample, pred):

    print(f"{msg} --> {'Spam' if p == 1 else 'Ham'}")
```

